

Engineering for electric mobility and energy storage
— from the motor controller's first boot to the app
in your pocket.

Jakub Szypicyn



Now

Head of Connected Systems, Skarper

Previously

Minimal, Breathe Battery Technologies, Skarper, WeCorp
Industries

Education

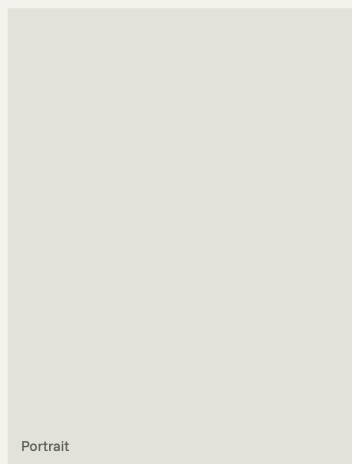
PhD, Imperial College London

Email ↗

LinkedIn ↗

CV ↓

About (draft)



I like products that move. For seven years I've taken e-bikes, battery labs and electric cargo vehicles from first boot to fleets in the field.

I design the boards, write the firmware, and build the pipelines that keep machines updated long after they leave the factory.

Before industry, a PhD at Imperial on reconfigurable analogue circuits. Away from the desk: cycling and touring, Formula 1, photography.

Journey

Five machines, all electric.

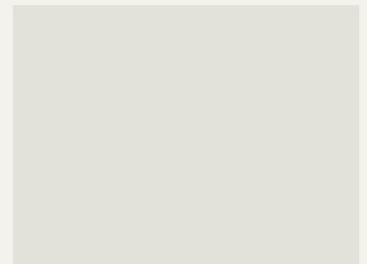


2020
WeCorp Industries, London

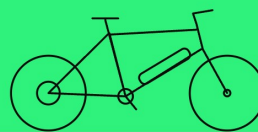
EEE Research Technical Lead

Built an STM32 drone-propeller test rig with C++ firmware and a Python DAQ GUI, and a field-oriented motor controller that cut power losses by 5%.

STM32 / C++ / FOC / Python DAQ



Propeller test rig

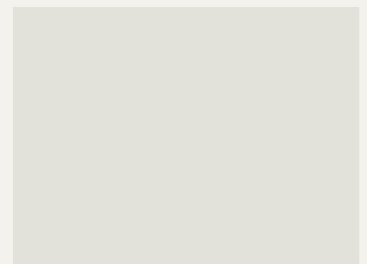


2021 – 2024
Skarper, London

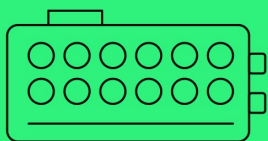
Senior Electronics Engineer

Second hire. Architected the 48V drive system — motor controller, VCU, UI and BMS — and wrote the first firmware for every board. Bring-up to mass production: 2,000+ units shipped, PCB BOM cost cut by more than half.

48V drives / BMS / EN 15194 / MISRA C / Production test



Drive system



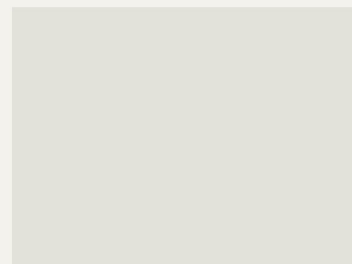
2024

2024 – 2025
Breathe Battery Technologies, London

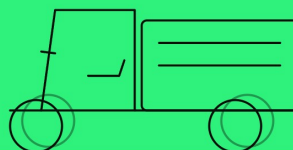
Embedded Firmware & Data Engineer

An Ethernet/TFTP bootloader for zero-touch updates across STM32 lab assets, CI/CD and Python services automating battery lab testing, and a Simscape pack model for BMS validation.

Bootloaders / GitLab CI / Azure / gRPC / Simscape



Battery lab



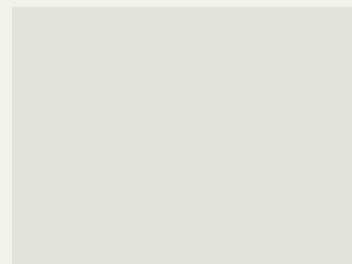
2025

2025 – 2026
Minimal, London

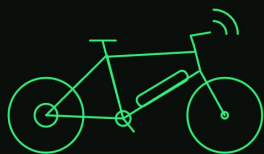
Senior Embedded Firmware & Data Engineer

A CAN bootloader and fleet OTA for 30+ connected vehicles. Firmware builds went from half a day to under ten minutes, releases from monthly to weekly.

CAN / Fleet OTA / BalenaOS / MISRA C / EN ISO 13849-1



Cargo fleet

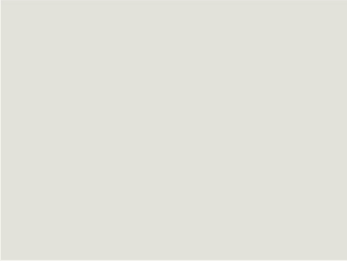


2026 – Now
Skarper, London

Head of Connected Systems

Rejoined to lead engineering: secure OTA with MCUboot and signed A/B images, a connected product platform from firmware to app, product cybersecurity and the EU Battery Passport.

MCUboot / BLE GATT / Supabase / Expo / EU Battery Passport



Today

Stack

Silicon to cloud.

05	App & cloud	Supabase, PostgreSQL, Cloudflare, Expo / EAS, AWS, Azure, FastAPI
04	OTA & release	MCUboot, signed A/B images with rollback, CAN and TFTP bootloaders, GitHub Actions, GitLab CI
03	Firmware	C, C++, STM32, Nordic nRF52, FreeRTOS, ChibiOS, bare metal
02	Interfaces	CAN, BLE with custom GATT profiles, SPI, I2C, Ethernet, MQTT
01	Hardware	PCB design and bring-up, 48V motor drives, FOC, BMS, production test
Compliance	Standards	EN 15194, EN ISO 13849-1, EN 17860, MISRA C:2023, EU Battery Passport, product cybersecurity

PhD

Circuits & Systems

Imperial College London, 2017 – 2021

Thesis on memristor-enabled reconfigurable analogue systems. Graduate Teaching Assistant in EE labs, amplifier design and FPGA.

MEng

Electronic Engineering, First Class

Imperial College London, 2013 – 2017

85%, top 10% of cohort. Dean's List and the Nicholas Battersby Prize for analogue electronics.

Patents & publications

Patent, 2022	US11421527B2, co-inventor	
IEEE MEMRISYS	[Paper title]	
IEEE CAS	[Paper title]	
IEEE	[Paper title]	

Contact

Let's build something that moves.

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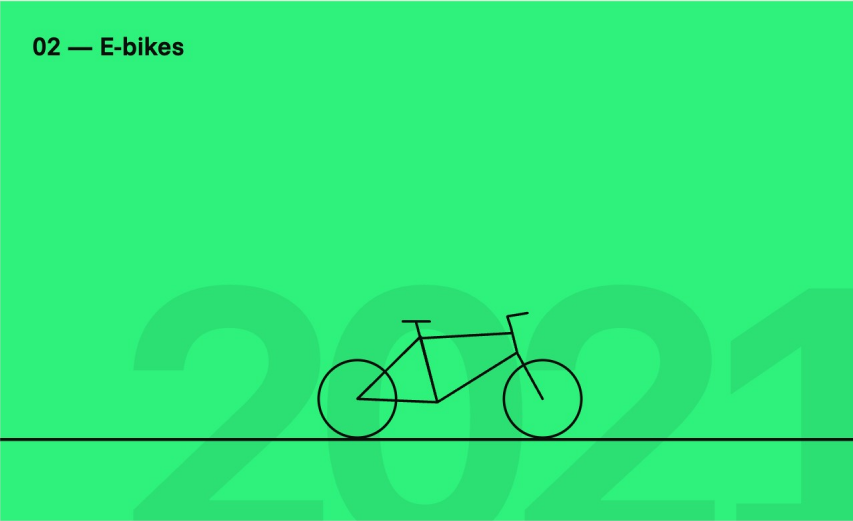
Page break, scrolled through

Band pins. Machine rides. Year drifts the other way.



0% · Enter

Band pins to the viewport. The bike rolls in from the left edge.



50% · Ride

Bike tracks the scroll, wheels turn. The year slides slower, the opposite way.



100% · Exit

Bike rides off right, the band unpins and the role text rises in line by line.

HEAD OF CONNECTED SYSTEMS • LONDON

Jakub Szypicyn.

I build connected hardware end to end — firmware, secure OTA, cloud and the app in your pocket.

[Email me](#)[LinkedIn](#)

2,000+
UNITS SHIPPED

PhD
IMPERIAL

02 — JOURNEY

Five machines, one thread.

CH. 02 / 05 — E-BIKES

2021-24



SKARPER • LONDON

Senior Electronics Engineer

JAN 2021 — SEP 2024 • EMPLOYEE #2

Architected the 48V drive system — motor controller, VCU, UI and BMS — and wrote the first firmware for every board. 2,000+ units shipped.

[48V drives](#)[EN 15194](#)[MISRA C](#)

[PHOTO — DRIVE SYSTEM]

CH. 03 / 05 — BATTERIES

2024-25